The Effectiveness of Monetary Policy in the context of COVID-19: A Survey of the Literature Review

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Abstract

The COVID-19 pandemic caused many economic and financial disruptions in several countries. This outbreak induced many economies to implement expansionary monetary policies to help alleviate the economic effects of the crisis. While many countries resorted to the conventional monetary tools, others, mainly those that have effective zero lower bound interest rate, implemented unconventional tools such as forward guidance and quantitative easing. The aim of this paper is to conduct a survey of the recent literature that assesses, empirically, the effectiveness of both conventional and unconventional monetary policies in the context of COVID-19 pandemic. The first section presents, theoretically, the transmission channels through which the monetary policies are transmitted to the economy. The second section presents the empirical literature on the topic. Overall, the evidence on conventional monetary policy tends more towards that the uncertainty of the pandemic lessened the effectiveness of the channels through which the policy operates. The effectiveness of unconventional monetary policy only affects the economy via the bond yields and stock prices and is contingent on the commitment and capacity of central banks to intervene. The results suggest further exploration of conventional and unconventional policy effects in uncertain circumstances to examine factors that enhance the efficiency of conventional monetary tools as well as the capacity and commitment of central banks in enhancing unconventional policy frameworks.

1. Introduction

The COVID-19 pandemic has not only posed a supreme global health crisis but has also triggered an economic downturn, posing challenges to both health and economic policy-makers. The damage brought by the outbreak has already surpassed the repercussions witnessed during the 2008 global financial crisis (Harvey, 2020). The policies aiming at combating the spread of the disease, such as the lockdown and remoteness, induced deteriorating impacts on the economic activity. Governments and central banks intervened in many countries to alleviate the economic impact of the crisis. Governments responded to the COVID-19 pandemic through various fiscal packages, such as direct cash transfers, unemployment support, healthcare funding, business support programs and infrastructure spending, to provide immediate relief to individuals and businesses and foster economic activity. Moreover, central banks around the world supported these efforts by implementing a variety of monetary policy tools to help prevent the whole crash of the economies. To provide monetary stimuli, many countries adopted conventional and/or unconventional monetary policies. Some economies, such as UK, Mexico, India, Brazil and Chile, lowered their policy rates, others, such as Brazil, Philippines, the United States and China, adjusted their foreign required reserve ratio, the US implemented unlimited quantitative easing and the Eurozone implemented new targeted long-term refinancing operations (Ozili and Arun, 2023). During this prolonged outbreak, coupled with frequent adjustments in monetary policy, it would be a distinctive opportunity to survey the literature on the effectiveness of monetary policy transmission to the economy.

Monetary transmission refers to the process through which changes in the central bank's monetary policy which include changes in interest rates or money supply influence economic growth, prices, and other economic factors. The transmission mechanism includes many stages and is characterized by long, variable and uncertain time lags, making it challenging to assess its effect on the economy (Di Maria, 2022). Yet, it is indispensable to understand the process to achieve macroeconomic targets such as full employment, economic resilience and price stability. Central banks, guided by theory, empirical evidence and experience, adjust their policies to achieve these targets.

In the context of conventional monetary policy, the study of monetary transmission mechanism gained a heightened focus in the mid-1990s. Mishkin (1995) played a pioneering role in investigating the monetary transmission process, emphasizing the primary channels of monetary and credit and their effect on the behavior of economic agents. Taylor (1995) introduced the notions of the interest rate channel and the exchange rate channel, while Ireland (2006) initiated extensive exploration into the asset price channel.

In exceptional circumstances, such as COVID-19, unconventional monetary policy tools become more appealing as the policy rate becomes close to the effective lower bound (zero). Some studies document a large degree of uncertainty during the Covid-19 pandemic (Altig et al., 2020; Coibion et al., 2020). Monetary policy might be less effective following a financial crisis since the latter

typically results from significant financial imbalances, such as substantial growth in private credit and a housing market boom, that precede them (Jorda et al., 2020). Borio (2014) finds that in times of uncertainty important transmission channels, such as the interest rate or credit channels, may encounter challenges.

Economic development is achieved by developing robust theories supported by empirical data. This paper starts with discussing the transmission channels through which conventional and unconventional monetary policies function. This is then followed by a survey on the empirical literature that assessed the effectiveness of different monetary transmission channels in the context of COVID-19. Eventually, the last section provides a brief concluding discussion.

2. Theoretical Part

In this section, the transmission channels of conventional and unconventional monetary policies.

2.1. Transmission Channels of Conventional Monetary Policy

It is important first to analyze, theoretically, how a (conventional) monetary policy channel is transmitted to the economy via four distinct but interconnected channels: interest rate channel, credit channel, exchange rate channel, and asset price and wealth channel (Mishkin, 1996).

2.1.1. Interest rate channel

The interest rate channel is regarded by many as the main channel of monetary policy transmission (Taylor, 1995; Loayza et al., 2002). It lies at the core of monetary transmission channel theory and the traditional Keynesian textbook IS-LM model, developed by Hicks in 1937. This policy action influences the individuals and business decision making units to change saving and consumption decisions, thus, affecting aggregate output and prices in the economy. Mishkin (1995) characterizes this channel as follows. A contractionary monetary policy leading to an increase in the interest rate causes the cost of borrowing to increase, thus causing consumption, investment and aggregate output to fall. On the contrary, following an expansionary policy, an increase in aggregate consumption and investment translates into higher output and eventually increases price levels due to the surge in the aggregate demand.

Factors that affect the effectiveness of this channel include degree of competition in the banking sector and the accessibility of alternative domestic funding sources (Brownbridge et al., 2017). The more the competitive the banking sector, the more rapid the effect of the change in the interest rate on the lending and deposit rates. Moreover, the availability of alternative sources of financing, such as bonds and stocks for households and firms, limits the monopolistic power of banks and thus, enhances the responsive of the interest rates.

2.1.2. Credit channel

Several theoretical papers discussed the existence of credit channel (Kashyap and Stein, 1995; Bemanke and Blinder, 1988). This channel describes how a change in the monetary policy affects the supply of credit, regardless of or in addition to a change in the interest rates. This channel operates through the banking system and financial markets, thus, providing insights into how changes in interest rates can affect the behavior of both lenders (banks) and borrowers (households and firms). Two channels exist here: the bank lending channel and the balance sheet channel. The former looks at the effect of monetary policy on a bank's capacity to lend firms, while the latter looks at the firm's capacity to borrow from markets in response to changes in their net worth following a monetary policy action (Kashyap and Stein, 1994; Hubbard, 1995).

The factor that most affects the effectiveness of this channel is the financial condition of a country's banking system. The weaker the financial system, reflected by low asset ratios or in terms of asymmetries of information and limited enforceability of contracts, the less banks tend to respond to monetary policy changes.

The channel suggests that an expansionary monetary policy enhances the share prices and net wealth of companies. Eventually, this decreases the moral hazard problems and enhances investment and aggregate output. Furthermore, a reduction in the interest rates lessens the business's cost of borrowing, thus increasing the cash flows of companies and contributing to more investments and economic expansion.

2.1.3. Exchange rate channel

This channel describes how a change in the monetary policy affects the exchange rate. It assumes a mutual correlation between interest and exchange rates. This influences the competitiveness of domestic goods and services as compared to foreign ones, thus, affecting the relative demand of both domestic and foreign goods and services[1].

Factors that most affect the effectiveness of this channel include the presence/absence of capital controls and the degree of substitutability between domestic and foreign assets (Kamin, 1998). In this context, the absence of capital controls enhances the effectiveness of monetary policy through this channel. Moreover, the larger the degree of substitutability between the domestic and foreign assets, the larger the effectiveness of the monetary policy as well (Dovciak, 1999).

An expansionary policy reduces the domestic interest rates. Following that, the interest rate differential between the domestic and foreign interest rates becomes less. Eventually, investors whose goal is to receive the highest return on investment may relocate capital investment to the foreign market, leading to an outflow of foreign capital. This reduces the demand of domestic currency in the foreign exchange market reducing its value. This depreciation decreases the price of exports and increases the price of imports, thus, enhancing net exports and aggregate

[1] https://www.elibrary.imf.org/view/book/9781589060944/ch06.xml

expenditure. Moreover, this depreciation may lead to higher prices of imported goods and services, contributing to inflation in the domestic economy.

2.1.4. Asset price and wealth channel

This channel describes how a change in the monetary policy affects asset prices and people's wealth. This channel operates by affecting the value of financial assets and, in turn, influencing consumer spending and business investment[2]. When central banks alter interest rates, it directly affects the returns on financial assets, such as bonds, stocks and real estate. Following this, changes in asset prices affect the overall wealth of households and businesses. Higher asset prices make individuals and businesses wealthier, thus increasing aggregate expenditure and output.

Factors affecting the effectiveness of the monetary policy via this channel include the level of development importance of bond, equity and real estate and the composition of financial portfolios (Kamins et al., 1998). More developed and diverse markets, along with a greater reliance on securities in financial portfolios enhance the sensitivity of the asset price channel to changes in monetary policy actions.

2.2. Transmission Channels of Unconventional Monetary Policy

In a low-interest-rate environment, where nominal interest rates are near or at zero, conventional monetary policy tools may lose effectiveness[3]. The zero lower bound leaves little or no room for additional cuts. This gives rise to the importance of employing a range of tools beyond traditional interest rate adjustments to support economic recovery. Amid the uncertainty brought by the pandemic, many central banks resorted to unconventional monetary policies such as forward guidance and quantitative easing (QE)[4] (Cantú and others, 2021). Forward guidance becomes one critical tool for signaling the central bank's commitment to keeping rates low for an extended period, providing additional stimulus to the economy and another critical tool to lower long-term interest rates is quantitative easing which involves large scale asset purchases programmes.

2.2.1. Forward Guidance

This strategy often involves signaling that interest rates will remain at low levels for an extended duration, thus, providing economic agents with assurance and guidance about the future trajectory of monetary policy. For instance, the Federal Reserve has explicitly stated that it intends to keep the federal funds rate within a target range of 0–0.25% until certain economic conditions are met. The criteria include achieving full employment and stabilize inflation to not exceed 2% for an extended period (Bernanke, 2022).

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[2] https://www.rba.gov.au/

[3] https://www.stlouisfed.org/from-the-president/speeches-and-presentations/2009/effective-monetary-policy-in-a-low-interest-rate-environment

[4] Very few countries (as Denmark, Japan and Switzerland) used negative interest rates

Forward guidance in policy practice can be classified into two types: Delphic guidance and Odyssey guidance (Campbell et al., 2012). The former involves the central bank offering future-related information to the market to help achieve policy plans, while the latter encompasses commitments from decision-making authorities about the future. Both types play vital roles when conventional monetary policies are ineffective in exceptional situations as the case of the pandemic (McKay et al., 2016).

In theory, this strategy may have a significant effect on the economy. Coenen and Warne (2013) used the ECB's New Area Wide Model and found that when nominal interest rates are at the effective ZLB, forward guidance can aid in price stability. However, real-world benefits may be less than predicted by standard models (Woodford, 2003). McKay et al. (2016) underscore that the pronounced theoretical impacts of forward guidance hinge significantly on the assumption of complete markets in these models.

2.2.2. Quantitative Easing

Given the constrained conventional monetary policy in many economies where policy rates approached their effective lower bound, many central banks resorted to large-scale Asset Purchase Programs (APPs) to stimulate output growth and fight price instability during the pandemic. Central banks purchase different assets, including long-maturity securities, such as government securities, securities from financial instituitions, corporate bonds, asset-packed securities,… to ensure liquidity and prevent insolvencies. This purchase of long-term securities helps fulfill demand from financial intermediaries, particularly during periods of heightened economic uncertainty when these securities are associated with high risk. Looking at the transmission channels through which this tool affects the economy, three channels can be highlighted: portfolio rebalancing effect, a liquidity effect and a signaling effect (Gern et al., 2015; Ulbrich and Bundesbank, 2016; Bernanke, 2020).

2.2.2.1.Portfolio Rebalancing Effect

This phenomenon is based on the notion that investors have distinct preferences for particular maturity periods. The monetary authority purchases bonds to alters the relative supply of various assets, and thus, affect their relative prices. Consequently, investors adjust their portfolios, causing the value of remaining government securities and their substitutes to increase. Following this, long term yields diminish and the term premium decreases.

2.2.2.2.Liquidity Effect

This effect is essential at initial stage of the crisis when central banks intervene and offer support to financial market (Ferdinandusse et al., 2020). During a crisis, economic agents lose trust in the asset values and financial stability of traders. This induces liquidity problem causing money markets to shrink and uncertainty to increase. When central banks provide quantitative easing by purchasing assets and implementing substantial APPs, confidence and liquidity can be restored.

2.2.2.3.Signaling Effect

This effect is related to forward guidance. When individuals view the central bank purchasing large scale assets, this is perceived as an extra commitment from the central bank to maintain short-term policy rates approach the effective ZLB for a prolonged duration. Following an anticipated decrease in future short-term interest rates, long-term interest rates decline too. This phenomenon is most pronounced as the recovery advances, inflation rises, and uncertainty regarding the future trajectory of interest rates intensifies.

These effects interact to enhance the overall financing conditions and foster aggregate demand. The fall in long-term interest rates facilitated by these effects enhance asset prices creating a wealth effect. This encourages individuals and businesses to spend more. Thus, the overall effect is to boost economic activity during the crisis.

3. Empirical Part

Assessing the empirical effectiveness of monetary policy is a crucial topic for research, and there is a huge body of work on this topic. In the context of COVID-19 pandemic that posed unprecedented and severe economic challenges, this topic gained more attention. Researchers employed various econometric models to investigate the different transmission channels of monetary policy, offering insights that guide policymakers in navigating the complexities of the current economic landscape. During the COVID-19 pandemic, central banks worldwide employed a mix of conventional and unconventional measures. In some situations, some central banks used interest rate changes to influence borrowing costs, spending, and investment in the economy. In other situations, central used additional unconventional tools to stimulate the economy when interest rates are at or near the effective lower bound.

This section demonstrates the effectiveness of conventional and unconventional monetary policies during COVID-19 crisis. We first provide empirical evidence on the different monetary transmission channels of conventional monetary policy. Then, we provide some evidence on the effectiveness of unconventional monetary tools.

3.1. Evidence on Conventional Monetary Policy

Although the literature on the efficiency of conventional monetary policy during the period of COVID-19, the strong evidence tends toward the ineffectiveness of most transmission channels.

The following studies found evidence on the effectiveness of certain transmission channels of conventional monetary policy (Loi and Dang, 2023; Ma et al., 2021; Yilmazkuday, 2022).

Loi and Dang (2023) assessed the bank lending channel of monetary transmission in Vietnam using the generalized method of moments (GMM). Using bank-level data within the period from 2007 to 2021, the authors find that the bank lending channel tends to be important in the period of the pandemic. The bank's loan supply is found to be sensitive to changes in the lending interest

rates. This suggests that the central bank can confidently continue utilizing traditional interest rate tools to conduct monetary policy in unfavorable circumstances. Based on the findings presented in this study, it can be asserted that the central bank can still exert influence on the economy through the banking system using conventional interest rate instruments.

Ma et al. (2022) analyzed the monetary policy shocks in a panel of 8 Asian economies in the context of COVID-19. They employed an ARDL model on data from 2005 to 2020. The results show that policy rate changes have an impact on output in both the short and long term, especially during recessions. In the short run, the response tends to be positive. However, in the long run, the output is diminished as the increase in interest rates leads private agents to retain their funds in interest-bearing avenues.

Yilmazkuday (2022) explores the impact of U.S. monetary policy, specifically policy rates, on exchange rates in 21 emerging-market countries during the outbreak. The author finds that a negative U.S. monetary policy shock resulted in the devaluation of currency in these countries. This suggests a new channel induced by disease outbreak. The result suggests that implementing policy measures in large economies as the U.S. can help combat the unanticipated economic repercussions of the COVID-19 pandemic in many countries.

Other studies found evidence on the ineffectiveness of certain transmission channels of conventional monetary policy during the pandemic (Yildirim and Turan, 2022; Mwamkonko, 2023; Mishchenko et al., 2021; Kurniawan and Astuti, 2023; Hariyanti et al., 2021).

Yildirim and Turan (2022) revisited the Interest rate channel in transmitting monetary policy shocks to the Turkish Economy. The study employs a threshold vector autoregressive (VAR) model in the period between 2006 and 2019. This channel was analyzed in the context of consumption and investment. The findings reveal that for the consumption transfer model, the interest rate channel seems effective only in high macroeconomic instability, while for the investment transfer model, the channel remains ineffective in both low and high macroeconomic instability scenarios. Thus, it is revealed that conventional monetary policy is an effective tool that can be used in high economic instability – case of COVID-19 – in the Turkish Economy.

In his study, Mwamkonko (2023) applied co-integration and VECM model to assess the effectiveness of five transmission channels in Tanzania. The author shows that the interest rate and asset price channels are ineffective in the economy. Though the credit channel is functional in Tanzania, it is weak. The exchange rate and anticipated inflation channels are the dominant transmission channels. This suggests enhancing the financial inclusion in Tanzania and necessitates the central bank's intervention to monitor inflation and exchange rates for being the dominant mechanisms.

Mishchenko et al. (2021) assessed the efficiency of the monetary transmission mechanism channels in Ukraine. The analysis reveals that the exchange rate channel is the dominant channel through which interest rate changes affect real output and inflation. The interest rate channel ranks

the second in terms of efficacy, while the credit channel is the least effective. The authors suggest that enhancing the efficiency of the monetary transmission channels in Ukraine requires to strengthen the institutional capacity of the central bank.

Kurniawan and Astuti (2023) examined the asset price and exchange rate channel in Indonesia. The results of the impulse response function (IRF) and variance decomposition analyses reveal that the effect of changes in the interest rate on interbank money market rates, stock prices and actual output is mainly through the asset price channel. The bond yield channel and exchange rate channels are relatively less effective. This is due to the small contribution of interest rates to volatility in the exchange rates and bond yields for Indonesia.

Hariyanti et al. (2021) employed an econometric model to study the effect of monetary policies adopted by the central bank of Indonesia during the pandemic on the economy's growth. Their results show that the conventional monetary policy based at decreasing interest rate to encourage spending did not work due to the indispensable risk associated with the pandemic. The insignificant relationship between interest rates and economic activity reflected the dominance of fiscal policy in influencing the aggregate Indonesian economy. Thus, the conventional monetary policy implemented in Indonesia doomed to failure.

Other studies emphasized the importance of the coordination between multiple policies, such as fiscal policies and conventional and unconventional monetary policies (Pinshi, 2020; Iyke and Maheepala, 2022; Prabheesh, 2021; Juhro et al., 2021).

Pinshi (2020) studied the effectiveness of cutting the interest rate in helping the economy of Congo to mitigate the effects of the pandemic. The results show that conventional monetary policy is not huge enough to encompass the pandemic. The interest rate channel failed to have an adequate positive outcome on inflation, the output gap and the exchange rate. This highlighted the importance of unconventional monetary tools where a good communication strategy and the testing of these tools are significant aspects in mitigating the economic damage brought by the outbreak. Moreover, the author emphasized on the importance of coordination between monetary and fiscal policies suggesting that a credible fiscal framework, accompanied by a credible monetary framework is indispensable in maintaining confidence, price stability and economic resilience.

Iyke and Maheepala (2022) studied the effect of conventional monetary policy in stock markets of some emerging countries in the context of COVID-19. By employing a model, they find that the emerging stock markets have experienced a decline in stock returns due to the adverse effects of the pandemic. However, the sole implementation of conventional monetary policy does not seem to alleviate the detrimental effects of the pandemic on these stock markets. The authors recommend implementing unconventional monetary policies (like forward guidance and quantitative easing), along with conventional monetary policies and fiscal procedures to give policy makers a chance to lift the stock markets. This is a strong evidence regarding the importance of considering unconventional monetary tools during the pandemic.

Using the ARDL and co-integration approach to investigate the effectiveness of monetary policy transmission in some emerging countries, Prabheesh (2021) found that in most economies, the transmission channel to inflation is ineffective due to uncertainty associated with the COVID-19 outbreak. In a very few economies, the transmission channel to output and credit stability is found to be effective. The results suggest that relying solely on conventional monetary policy may not be sufficient for managing macroeconomic stability during periods of heightened uncertainty. Instead, there is a need to reinforce the integration of other fiscal policies and monetary policies, such as forward guidance and quantitative easing, to enhance the effectiveness of monetary policy transmission (Juhro et al., 2021)

Feldkircher et al. (2021) employed a novel mixed frequency vector autoregressive (MF-VAR) model to investigate the effectiveness of US monetary policy during the COVID-19 recession. The results show that the monetary expansion caused by a shock in the policy rate and accompanied by quantitative easing enhances economic growth and stock market returns. Moreover, such a policy triggered the depreciation of the US dollar supporting external competitiveness of the economy. However, there was no significant effect on unemployment and inflation.

3.2. Evidence on unconventional monetary policy

The primary concern from the monetary policy perspective is the effectiveness of the monetary transmission mechanism, especially in the context of the zero-lower-bound. There is a significant debate on this issue, with some experts suggesting limited efficacy when interest rates are close to their lowest bound. Bernanke (2020) argues that unconventional monetary policy tools may provide more flexibility. Guerrieri et al. (2021) propose a combination of easing monetary policy and increasing social transfers to mitigate the effects of the outbreak. Unconventional tools such as equity injections, helicopter money, loan guarantees, and wage subsidies are seen as effective measures to maintain full employment and ensure financial and macroeconomic stability.

Some studies assessed the effectiveness of interest rate forward guidance on the economy in the context of uncertainty brought by the pandemic (Gerke et al., 2021; Levin and Sinha, 2020)

Gerke et al. (2021) assessed the effectiveness of unconventional monetary policies at the effective lower bound using a DSGE model for the euro area. In the context of forward guidance on interest rates, simulations show that the guidance has the potential to decrease macroeconomic volatility. The commitment, if deemed credible, can provide additional stimulus in unfavorable states and diminishes stimulus in favorable states. In the context of asset purchases, the simulations emphasize the indispensable impacts on economic output and inflation, solely when interest rates reach the effective lower bound. According to the simulations, unrestricted asset purchases at the effective lower bound can effectively stabilize inflation. However, the estimated asset purchase rule suggests that it cannot entirely offset the negative inflation bias, particularly when upper purchase limits constrain the potential purchase volume. The authors recommend an effective

reinvestment strategy which serves as a form of forward guidance to enhance the impact of asset purchases in the context of these upper limits.

Levin and Sinha (2020) provided some limits on the effectiveness of forward guidance policy in the context of COVID-19 outbreak. Based on a canonical New-Keynesian model, the authors highlight the limited foresight of policy makers in the early stages of the pandemic. The authors propose some arguments to address the "forward guidance puzzle". First, the central bank may have imperfect credibility in making longer-term commitments about the path of monetary policy. Second, the central bank may lack complete information of the right economic structure. Within this framework, the results suggest that delivering significant near-term monetary stimulus relies on making commitments that involve substantial output and inflation overshooting in the following years. Eventually, forward guidance may provide only inadequate overall paybacks and might even be counterproductive in certain cases.

The literature on the pandemic quantitative easing (QE) programs is rich with the majority of this literature focusing on the effect of these programs in the financial market, specifically on asset prices (Bernardini and De Nicola, 2020; Arora et al., 2021; Bernardini and Conti, 2021; Fratto et al., 2021; Hanlon, 2021).

All these studies reveal the temporal fading effect of quantitative easing (QE) on government bond yields. The real purchases lead to a temporary reduction in bond yields, typically lasting only a few days. Bernardini and Conti (2021) focused on comparing the effectiveness of ECB's announcements and actual purchases of assets using a VAR model. The results show a negative effect on bond yield and a positive effect on stock prices. However, the announcements provide the bulk of the effect. Fratto et al. (2021) and Ha and Hanlon (2021) investigated whether the characteristics of a country affect the relationship between QE programmes and financial assets. The former found that QE are more pronounced in countries with higher credibility, while the latter found that QE has a stronger effect in countries with less credibility. The two studies revealed that a larger scale of asset purchases is not accompanied by a more powerful effect on financial assets.

A limited number of studies examined the impact of QE on the macroeconomic factors as output and inflation (Akkaya et al., 2023; Garcia et al., 2022; Hertel et al., 2022).

Akkaya et al. (2023) employed a DSGE model in Sweden and found that with pandemic-era central bank asset purchases, GDP and inflation only increases by 0.1%. Using the same framework, Garcia et al. (2022) revealed that the effect in Chile was even less important. The results were more substantial in Poland for revealing that QE increased GDP and the price level by 0.6-0.7% (Hertel et al., 2022). Consequently, these studies reveal a relatively negligible effect of central bank asset purchases on output and inflation.

4. Concluding Discussion

Although there are mixed results on the monetary transmission mechanism of conventional monetary policy within the context of COVID-19, the evidence tends more towards that the uncertainty of the pandemic lessened the effectiveness of conventional monetary policy.

The use of unconventional policies in economies with effective zero bound, particularly forward guidance depends on the commitment of the central bank. In the context of quantitative easing, the major effect is reflected via the impacts on bond yields and stock prices, rather than other macroeconomic variables as output and inflation. The effect of QE varies significantly among countries. The effectiveness of QE is mainly attributed to announcement effects, particularly through signaling channels, rather than the actual purchase of assets. The most important factor lies in the central bank's capacity to intervene when needed.

These results impose some implications for future research work. The survey of the literature suggests further exploration of conventional and unconventional policy effects in uncertain circumstances, such as the pandemic. It is important to investigate the effectiveness of conventional monetary pools across different countries with different economic conditions and examine ways to enhance the efficiency of such tools during times of uncertainty. It is also important to examine factors that affect the capacity and commitment of central banks in enhancing unconventional policy frameworks. Moreover, investigating the interaction between conventional and unconventional policies is also crucial for future research. Future research work should assess how these policies complement or conflict with each other and identify optimal combinations for addressing economic challenges effectively.

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